

TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

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VERSION **v. 1.1**

RELATED SUPPORT

– TEMPLATE GUIDE Key Project Information & VPA Design Document v.1.1

This document contains the following Sections

Key Project Information

Section A – Description of project

Section B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

Section C – Duration and crediting period

Section D – Summary of Safeguarding Principles and Gender Sensitive Assessment

Section E – Summary of Local stakeholder consultation

Appendix 1 – Safeguarding Principles Assessment (mandatory)

Appendix 2 - Contact information of VPA Implementer (mandatory)

Appendix 3 - Summary of Approved Design Changes (project specific)

KEY PROJECT INFORMATION

GS ID of Project	GS11206
Title of Project	GS7591 VPA 47 Malawi Improved Cookstove Project
Time of First Submission Date	15/06/2021
Date of Design Certification	22/02/2023
Version number of the VPA-DD	10
Completion date of version	27/06/2023
Coordinating/managing entity	Amie Nevin – CO2balance UK Ltd
VPA Implementer (s)	United Purpose
Project Participants and any communities involved	CO2balance UK Ltd
Host Country (ies)	Republic of Malawi
GS ID and Title of applicable Design Certified VPA	N/A
GS ID and Title of applicable Performance Certified VPA	N/A
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☒ Small Scale ☐ Large Scale
Other Requirements applied	N/A
Methodology (ies) applied and version number	GS4GG Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC) v3.1
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular

☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact	Estimate d Annual Average	Units or Products
13 Climate Action (mandatory)	13.B.1: Number of least developed countries and small island developing States with nationally determined contributions, long-term strategies, national adaptation plans, strategies as reported in adaptation communications and national communications	Emissions Reductions	34,034	Emissions Reductions
3 Good Health and Well-Being	3.9: By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	Total reduction in household air pollution for project activity in year y (%)	31%	HAPR _y
5 Gender Equality	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location	Total reduction time spent collecting firewood and cooking for project activity in year y (%)	40%	TR _y
7 Affordable and Clean Energy	7.2: By 2030, increase substantially the share of renewable energy in the global energy mix	Number of project ICS active in year y	24,000	ICS _{a,y}

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

CO2balance UK Ltd (CO2balance) and United Purpose will distribute around 30,000 local stoves across Lilongwe and Chiradzulu Districts of Malawi, where rural households typically cook on inefficient open fires or three stone fires. The project stoves will generally be sold at local markets and via agents, also known as Last Mile Distributors (LMDs).

The purpose of the Project is to reduce greenhouse gas emissions at a household level through the distribution of improved cookstoves which reduce wood consumption. As well as reducing greenhouse gas emissions, the Project will provide various benefits to the end user households and communities whilst helping to meet the UN Sustainable Development Goals (SDG) targets. Benefits include improved health through reduction in indoor air pollution, saving women time on unpaid chores, economic development by creating manufacturing jobs and sales opportunities, and reducing the income expenditure on fuelwood for stove recipients.

A.1.1. Eligibility of the project under approved PoA

This Project is automatically eligible for Gold Standard Certification as it follows the approved Gold Standard Activity Requirements 1.2(b) End-Use Energy Efficiency: efficient cooking, and eligible under the PoA as it is in one of the defined host countries (section A.3), Republic of Malawi, and implements one of the defined project technologies (section A.4), improved cookstoves.

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
1a	Types of Project	Eligible Projects shall include physical action/implementation on the ground. Pre-identified eligible Project types are identified in the Eligibility	The projects involve the sale of locally designed and manufactured improved cookstoves.

		Principles and Requirements section.	The project type is eligible under Community Services Activity Requirements 3.1.1(b) End-Use Energy Efficiency: efficient cooking.
1b	Location of Project	Projects may be located in any part of the world.	The VPA is located in in Malawi as specified in Section A.2 of this document. Malawi is one of the countries outlined in the PoA-DD. The VPA is uniquely defined by a range of GPS coordinates and current administrative maps to define the project boundary.
1c	Project Area, Project Boundary and Scale	The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. In order to avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the project shall demonstrate that there is no double counting of impacts at design and	The project boundary, including GPS co-ordinates and maps are included in Section A.2 and B.3. This VPA is not included by any other carbon standard and will be capped at an energy output of 60Gwh (180GWth) per year as per CDM small-scale requirements. The host country, Malawi, and location of each VPA is specified in Section A.2, in line with the locations outlined in Section A.3 of the PoA-DD. The project boundary, including GPS co-ordinates and maps are included in Sections A.2 and B.3. This VPA is not included by any other carbon standard and will be capped at an energy output of 60Gwh (180GWth) per year as per CDM small-scale requirements.

		performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects).	
1d	Host Country Requirements	Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	The Project Activity does not violate any of the Host Country's legal, environmental, ecological and social regulations. The Project Activity <i>reduces</i> greenhouse gas emissions, which is inline with the Environment Management Act, 2017 ¹ ; and the reduction of gender inequality through sales and monitoring through Womens' Groups is inline with the National Policy on Gender and Development ² .
1e	Contact Details	As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the	Contact details of the Project Developer are included in Appendix 1.

¹ Malawi Environment Management Act, No. 19 of 2017, Parliament of Malawi, 2017. <http://extwprlegs1.fao.org/docs/pdf/mlw169354.pdf>

² National Gender Policy on Gender and Development, Second Edition, Ministry of Gender, Children, Disability AND Social Welfare, 2015. <https://npc.mw/wp-content/uploads/2020/07/National-Gender-Policy-2015.pdf>

		required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.
1f	Legal Ownership	Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising. At the point of sale, purchasers of the stove are informed of the transfer of carbon rights in exchange for a stove subsidy. The terms of the carbon transfer are stated on a warranty card received at the time of purchase.
1g	Other Rights	As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for There are no disputes or contested rights that have been identified in relation to rights relevant to the project activity

		example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further Project implementation in affected areas.	
1h	Official Development Assistance (ODA) Declaration	All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification.	A declaration confirming that there is no diversion of ODA has been submitted.
2	Community Service Activity Requirements	2.1.2) All CSA projects shall lead to climate change mitigation and/or adaption by providing or improving access to services/resources at household or community or institution level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc. 2.1.3) In relation to the above all Projects shall	By distributing improved cookstoves the cookstove projects will ensure that households consume less firewood during the process of domestic cooking. As a result there shall be a reduction of carbon dioxide emissions from the combustion process at household level. This mitigates climate change by increasing access to improved cooking technologies amongst rural communities

	therefore confirm to Gold Standard for the Global Goals Principles & Requirements (and associated documents)	The project conforms with GS4GG Principles and Requirements. As set out in the VPA-DD and PDD.
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A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

In this project CO2balance, is the managing entity and is responsible for communication with the Gold Standard. CO2balance's implementing partner is United Purpose who manufacture the stoves and market them using Last Mile Distributors (LMDs) to sell the stoves. UP/LMDs conduct the monitoring. CO2balance own the legal rights to the carbon credits created by the project.

The purchasers of the project stove are informed of the carbon transfer at the point of sale and receive a receipt which confirms the transfer of carbon rights. This entitles CO2balance to the rights of any carbon saved by the project.

A.2. Location of project

Projects are located in the districts of Lilongwe Rural and Chiradzulu, Malawi (see maps below).

Projects are operated from United Purpose Malawi's head office in Green Heritage House, First Floor, Area 13, PO Box 159, Lilongwe, Malawi.

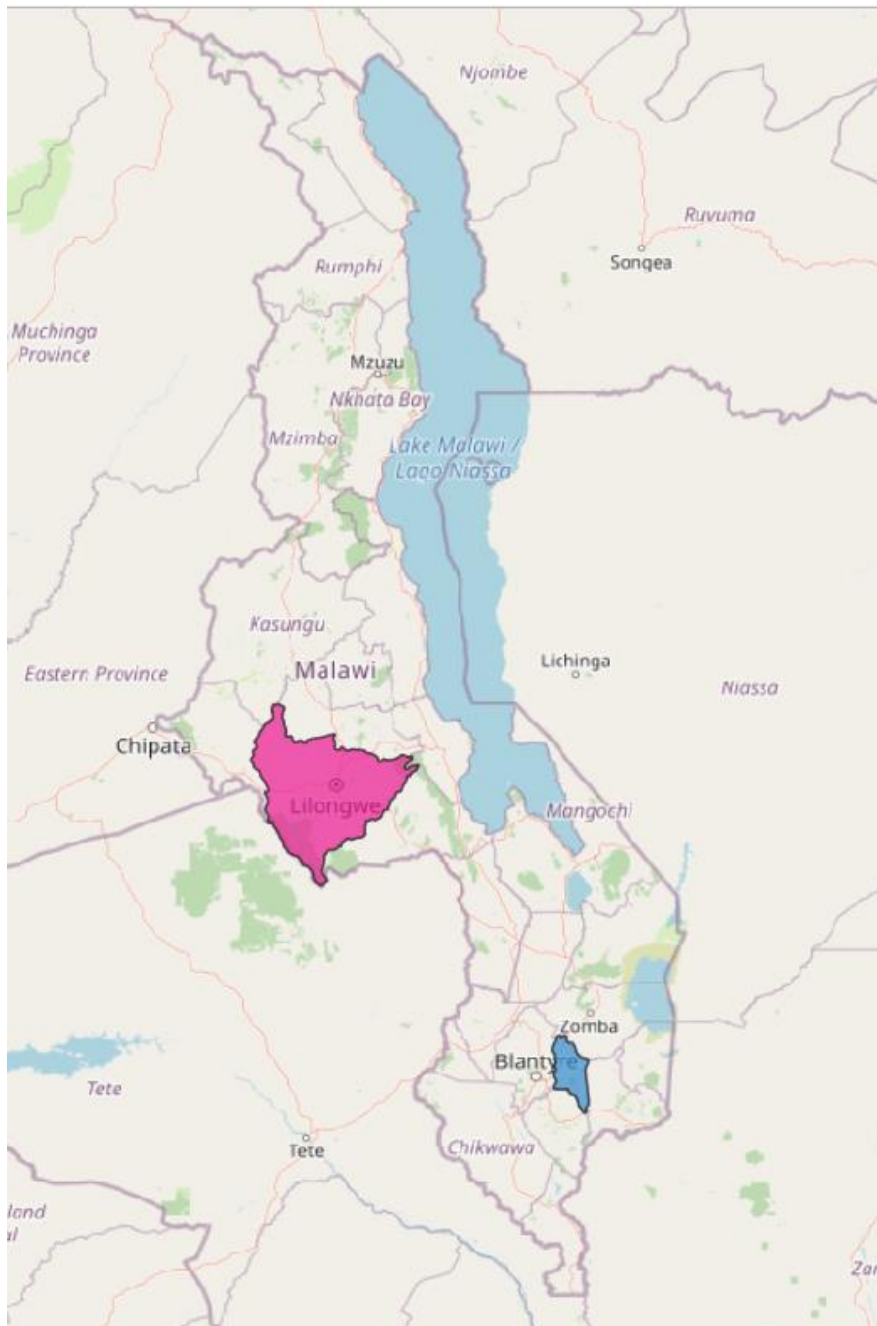


Figure 1. Project boundary within Republic of Malawi.



Figure 2. The project boundary in greater detail.

	Latitude	Longitude
North	-13.48179	33.41085
East	-15.68102	35.32668
South	-16.00360	35.29823
West	-13.48179	33.24500

A.3. Technologies and/or measures

Households in rural areas of the mountainous northern regions typically cook on traditional open fires and 3- stone stoves. These are extremely inefficient and demand a lot of firewood, resulting in high CO2 emissions, deforestation and exposure to indoor air pollution.

The project stoves will displace the inefficient baseline technology, and reduce CO2 emissions by reducing wooduse by roughly 45%. The reduced use and demand for firewood will reduce CO2 emissions, reduce demand of wood collection on local forests (for those that collect wood), reduce requirement to purchase wood (for those who buy wood), reduce the time spent collecting wood and cooking, and reduce exposure to harmful indoor air pollution.

In the cookstove sector, it is generally accepted that market-based approaches are the most effective as they avoid distorting the market through subsidies and free hand-outs. However, in remote rural areas, this becomes challenging as a purely commercial approach often fails when extremely poor people or vulnerable groups are targeted.

To combat this problem, United Purpose has established a stove value chain in the target districts Lilongwe Rural and Chiradzulu, which will help to reach remote rural households, which would not normally have access to these technologies.

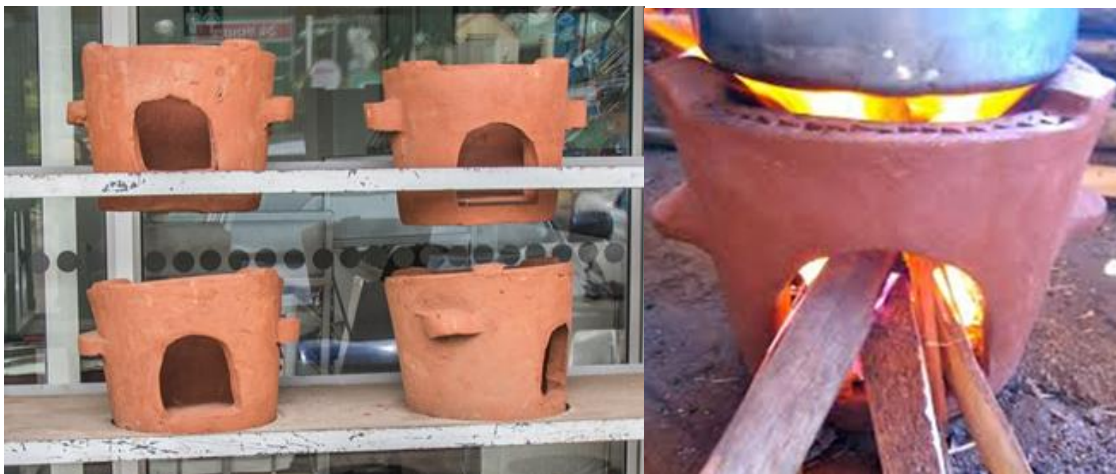
More specifically, United Purpose will:

- Establish or strengthen stove production groups within each district
- Create new value chains by working with local traders and promoters as distribution points
- Distribute free stoves to 'Social Cash Transfer' households using a voucher system
- Support sales through sales agents

'Social Cash Transfer' households refer to labour constrained families which are within the poorest 10% of society. Whilst supporting the most vulnerable people, this method increases demand and interest for these stoves from the wider community.

All stoves are distributed by trained sales agents who provide an explanation on how to use and care for the stove. A simple user and care manual is included with each stove that the beneficiary can keep at home. If users experience a problem with their stove they can contact their local sales agent, or call the number of the emission reduction receipt to directly reach United Purpose.

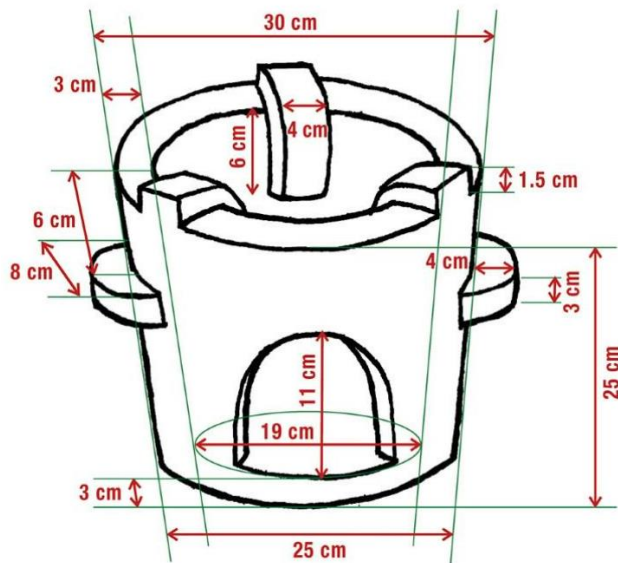
Production is undertaken by trained community members using a metal paddle mould to standardise the shape, ensuring the desired thermal efficiency is achieved.



The project technology is a locally produced improved clay cookstove, known as the Chitetezo Mbaula, a portable clay stove. The Chitetezo Mbaula is one of the most widely dispersed cleaner cookstoves in Malawi and was originally designed by the GIZ supported ProBEC programme. 'This is a portable stove made from clay that has been designed to make cooking easier, faster and safer for the user, whilst using less wood fuel for cooking'³ using 53% of the fuel that a three stone fire requires⁴.

³ <https://cepa.org.mw/Library/climate-change/Chitetezo%20Mbaula%20Guidelines.pdf/view>

⁴ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5930929/pdf/nihms905929.pdf>



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Chitetezo Mbaula means 'protecting stove' in Chichewa due to its safe and stable design. The fire is contained in a chamber with an opening at the front which wood is fed into. It is able to heat one pot on the top. The stove has an estimated lifetime of at least 3 years based on a report conducted by GIZ. Each stove has a unique serial number to distinguish it from other stoves and avoid double counting.

The performance of this stove

Lifetime: 3.7 years⁶

Thermal efficiency: 22.6%⁷

A.4. Scale of the project

This is a small scale “type II” project as it focuses on “energy efficiency improvement project activities” and must not exceed the equivalent of 60 gigawatt hours per year every year. VERs are capped at 60,000 per year.

⁵ Quality control for Chitetezo Mbaula, supplementary document provided

⁶ Lifespan Survey for Chitetezo Mbaula Stoves, Energising Development, Malawi. Deutsche Gesellschaft für Internationalen Zusammenarbeit (GIZ) GmbH.

⁷ Centre for Research in Energy and Energy Conservation, Preliminary Stove Performance Test Results, Report No. B/TR/2020/005

A.5. Funding sources of project

The project is funded by private funding through United Purpose.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

Gold Standard Methodology: Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC) v3.1.

B.2. Applicability of methodology (ies)

The project is expressly applicable under section 1 of TPDDTEC v3.1.

Methodology Requirement	Project
1. 'The project boundary needs to be clearly identified, and the technologies counted in the project are not included in another voluntary market or CDM project activity. In some cases there maybe another similar activity within the same target area. Project proponents must therefore have a survey mechanism in place together with appropriate mitigation measures so as to prevent any possibility of double counting'	The project area, the districts of Lilongwe and Chiradzulu, have been clearly demarcated using political boundaries recognized in Malawi. Each technology will be individually tagged with an identification code which is stored securely in the project database. Regular project surveys together with distribution records will ensure that the technologies included in the project are not double counted. Additionally, the project partner is a major contributor to this sector in Malawi, and so are aware of developments of additional projects. By project design, stoves will be distributed in different communities than other projects within the same project area.

2. Technologies have a continuous useful energy output of less than 150kW per unit (defined as total energy delivered usefully from start to end of operation of a unit divided by time of operation). For technologies or practices that do not deliver thermal energy in the project scenario but only displace thermal energy supplied in the baseline scenario, the 150kW threshold applies to the displaced baseline technology.	The estimated energy output of the project stove is expected to be no more than 3 kW which is well within the methodological limit of 150kW. See latest version of GS11206 ETP Stoves Ex-Antes Tab Ex-Antes (2020)!J22 for calculation.
3. The use of the baseline technology as a backup or auxiliary technology in parallel with the improved technology introduced by the project activity is permitted as long as a mechanism is put into place to encourage the removal of the old technology and the definitive discontinuity of its use.	As noted in the Gold Standard Methodology p.5. 'the removal and continued non-use of three stone fires and other easily constructed traditional devices (the baseline technology replaced by this project activity) is in many cases unlikely and impractical to monitor.' The project stove is purchased by users as they see some benefit in using it over their current woodfuel, such as saving on wood costs, saving time on wood collection, ability to move the stove, ease of use, and/or reduced smoke. Therefore, households are unlikely to use the baseline stoves if they have invested in the project stove.
a) The project documentation must provide a clear description of the approach chosen and the monitoring plan must allow for a good understanding of the extent to which the baseline technology is still in use after the introduction of the	Overall use of the baseline technology will be monitored in conjunction with that of the project technology, as will the emergence of any other baseline technology by targeted end users. As per the Methodology kitchen surveys will be

improved technology, whether the existing baseline technology is not surrendered at the time of the introduction of the improved technology, or whether a new baseline technology is acquired and put to use by targeted end users during the project crediting period.	carried out at regular intervals to determine any changes in baseline technology use.
b)'The success of the mechanism put into place must therefore be monitored, and the approach must be adjusted if proven unsuccessful. If an old technology remains in use in parallel with the improved technology, corresponding emissions must of course be accounted for as part of the project emissions.'	Parallel baseline technology use will be revealed during monitoring and its effect on emissions reductions will be captured in the KPT and usage surveys. The uptake rate U will also be determined by surveys and hence used to account for parallel baseline and project technology use.
4. The project proponent must clearly communicate to all project participants the entity that is claiming ownership rights of and selling the emission reductions resulting from the project activity. This must be communicated to the technology producers and the retailers of the improved technology or the renewable fuel in use in the project situation by contract or clear written assertions in the transaction paperwork. If the claimants are not the project technology end users, the end users should be notified that they cannot claim for emission reductions from the project.	Transfer of carbon rights is included in the receipt/warranty provided to stove recipients at the point of sale. An initial was conducted remotely (please see SCR) in which the carbon transfer rights were explained. This was then repeated in an in person LSC.
5. Project activities making use of a new biomass feedstock in the project situation (e.g. shift from non-renewable to green	As the technology used in this project has been specifically designed to displace baseline feedstock use via fuelwood,

charcoal, plant oil or renewable biomass briquettes) must comply with relevant Gold Standard specific requirements for biomass related project activities, as defined in the latest version of the Gold Standard rules.	rather than a new biomass feedstock, this criterion is not applicable to this project. The emission reductions from this project will result from a change in quantity of fuel consumed, rather than change of fuel type.
a) Adequate evidence is supplied to demonstrate that indoor air pollution (IAP) levels are not worsened compared to the baseline, and greenhouse gases (as listed in section 2.1) emitted by the project fuel/stove combination are estimated with adequate precision. The project fuel/stove combination may include instances in which the project stove is a baseline stove.	The fuel used in both the project and baseline scenario is the same, so as a result of the increase in efficiency of the stoves, IAP should be reduced by 31%.
b) Records of renewable fuel sales may not be used as sole parameters for emission reduction calculation, but may be used as data informing the equations in section 2.0 of this methodology if correlated to data on distribution and results of field tests and surveys confirming (a) actual use of the renewable fuel and usage patterns such as average fraction of non-renewable fuels used in mixed combustion or seasonal variation of fuel types, (b) GHG emissions, (c) evidence of CO levels not deteriorating (d) any further factors effecting emission reductions significantly.	Renewable fuels are not sold as part of this project therefore this point is not applicable.

B.3. Project boundary

Source		GHGs Included?		Justification/Explanation
Baseline scenario	Combustion of firewood	CO ₂	Yes	Important source of emissions
		CH ₄	Yes	Important source of emissions
		N ₂ O	Yes	Can be significant for some fuels
Project scenario	Combustion of firewood	CO ₂	Yes	Important source of emissions
		CH ₄	Yes	Important source of emissions
		N ₂ O	Yes	Can be significant for some fuels

B.4. Establishment and description of baseline scenario

In Lilongwe Rural and Chiradzulu, Malawi, local people typically use wood fuel on inefficient three stone fires for cooking. This process results in the release of greenhouse gas emissions from the combustion of wood. This can be reduced if households have access to improved cooking solutions. Households also use a minimal amount of charcoal, typically for making tea and coffee.

Co2balance seeks to register this project as a Gold Standard small scale project using the methodology "Technologies and Practices to Displace Decentralized Thermal Energy Consumption Version v3.1." In this project CO2balance will work with United Purpose (an NGO), local communities, and local government in central and southern regions to distribute improved cookstoves, reducing the woodfuel demand within households.

The baseline situation is not expected to change significantly during the next years considering the current situation in Malawi, its economic development of the last years and predictions for the future. Malawi is an LDC and is one of the poorest countries in the world with a Human Development Index ranking of 163 out of 191 countries worldwide in 2021⁸.

Baseline Scenario:

The baseline scenario is assessed through use of:

- Baseline Kitchen Survey
- Application of the methodology default value to inform baseline fuel consumption.

⁸ https://www.theglobaleconomy.com/rankings/human_development/

In accordance with the GS4GG Methodology “Technologies and Practices to Displace Decentralized Thermal Energy Consumption v3.1” (TPDDTEC), the baseline survey is carried out using representative and random sampling. The sample size is determined in line with the methodological minimum sample size and confidence requirements.

The baseline study was conducted between 02/09/2021 and 08/09/2021 in line with GS4GG *Principles & Requirements* and the TPDDTEC v3.1 methodology. There were 241 randomly selected interviewees, with at least 120 in each Project District (120 in Chiradzulu, 121 in Lilongwe Rural).

From the baseline survey, following season adjustment, it was determined that 86% of Stove Usage was using Three Stone Fires, 9.09% wood ICS Devices and 4.61% charcoal stoves. All remaining types De Minimis begin less than 5%.

For the purpose of the baseline calculations, the wood ICS and charcoal stoves have been combined to represent a 14% use of improved devices in the baseline scenario. This is considered conservative, rather than disregarding the charcoal stoves as De Minimis, both devices have the default efficiency of 0.2, and fuelwood has a higher total emissions per unit of energy.

No suppressed demand is considered in the baseline scenario of this project.

This project applies the TPDDTEC v3.1 default to determine baseline fuel consumption. In line with the methodology, this is a viable option which ensures conservativeness. The default value of 0.5 tonnes per capita per year was applied to determine the wood fuel consumption. In accordance with the Ex-Ante calculations, a default value of 6.03 kg/HH/day has been calculated using national statistics available from the 2018 Malawi Population and Household Census report (pages 31 and 32) on average household sizes as stated below:

Default fuel use (T/pp/yr) x household size (persons) = Baseline fuel use (kg/HH/day)
 $(0.5 \times 4.4 \times 1000) / 365 = 6.02739726$.

As the project technology is installed at the start of the project, the baseline scenario is considered fixed throughout the crediting period.

B.5. Demonstration of additionality

The table below is only applicable if the proposed project is deemed additional, as defined by the applied approved methodology or activity requirement or product requirement.

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version	As demonstrated in the Gold Standard for the Global Goals Community Services Activity Requirements section 4.1.9 - Projects that meet any of the following
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number and the specific paragraph, if applicable).

criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification:

1. (a) Positive list (Annex B)
2. (b) Projects located in LDC, SIDS, LLDC
3. (c) Micro-scale projects

Describe how the proposed project meets the criteria for deemed additionality.

The project meets criterion 1. Positive list, Annex B of the Community Services Activity Requirements. Section 1.1.3 of Annex B states.

"Project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where each unit results in ≤ 600 MWh of energy savings per year or ≤ 600 tonnes of emission reductions per year".

Ex ante calculations have shown that emission reductions per improved cookstove per year come to 0.67 tCO₂e.

The project is therefore additional.

B.5.1. Prior Consideration

Date of First Submission (15/06/2021) is a year from Start Date (16/06/2020).

B.5.2. Ongoing Financial Need

N/A

B.6. Sustainable Development Goals (SDG) outcomes

SDG Impact

Sustainable Development Goals Targeted	Most relevant SDG Target	Indicator (Proposed or SDG Indicator)
SDG 3. – Good Health and Well Being	3.9: By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	HAPR _y : total reduction in household air pollution for project activity in year y (%)
SDG 5. – Gender Equality	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location	TR _y : total reduction time spent collecting firewood and cooking for project activity in year y (%)
SDG 7. – Affordable and Clean Energy	7.2: By 2030, increase substantially the share of renewable energy in the global energy mix	ICS _{a,y} : number of project ICS active in year y
SDG 13. - Climate Action (mandatory)	13.B.1: Number of least developed countries and small island developing States with nationally determined contributions, long-term strategies, national adaptation plans, strategies as reported in adaptation communications and national communications	Emissions Reductions

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

SDG 3 (Good Health and Wellbeing)

Switching from an unimproved stove to an improved cookstove reduces the volume of non-renewable biomass that needs to be burned for the household's daily cooking needs. Therefore, the reduction in non-renewable biomass burned can be taken as a proxy indicator of how these VPAs contribute towards the SDG 3 target of substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.

The average difference in fuel use per technology per day in the project will be subtracted from the baseline figure which comes from the Kitchen Performance test and is fixed throughout the crediting period. This difference will then be used to calculate the % decrease in fuel use, which is the figure used to calculate the VPAs' contribution to the SDG target.

The overall reduction in HAP by the project activity are then calculated as follows:

$$\text{HAPR}_y = ((\text{Pb},y - \text{Pp},y) / \text{Pb},y) * \text{Up},y$$

Where:

HAPR _y	Total reduction in Household Air Pollution for project activity in year y (%)
Pb, _y	Quantity of fuel that is consumed in the baseline scenario b during year y (kg/household-day)
Pp, _y	Quantity of fuel that is consumed in the project scenario p during year y (kg/household-day)
Up, _y	Usage rate in project scenario p during year y

SDG 5 (Gender Equality)

Women and girls perform the majority of unpaid domestic work⁹. This leaves them with less time to rest, study and realise their economic potential, leaving them in time poverty. In regard to time, women are poorer than men as unpaid domestic duties, such as collecting firewood and cooking, must be added to their market productive work, making time much scarcer¹⁰. Women are widely recognised as being principally responsible for natural resource collection¹¹.

⁹ UN (2017) 'Progress towards the Sustainable Development Goals (E/2017/66)'. Available at <https://unstats.un.org/sdgs/files/report/2017/secretary-general-sdg-report-2017--EN.pdf>

¹⁰ Charmes, J 'A Review of Empirical Evidence on Time Use in Africa from UN-Sponsored Surveys', in World Bank (2006) 'Gender, Times Use, and Poverty in Sub-Saharan Africa'. World Bank Working Paper No. 73

¹¹ Nankhuni (2004) 'Environmental Degradation, Resource Scarcity and Children's Welfare in Malawi: School Attendance, School Progress, and Children's Health'

These trends demonstrate that reducing the amount of firewood required by households and access to a more efficient cooking technology has the potential to reduce the time poverty of women, because the time burden of collecting firewood and cooking, which falls disproportionately on women, will be reduced. The decrease per household in time spent gathering firewood and cooking will be taken as a proxy contribution towards the SDG target.

The overall reduction in time spent collecting firewood and cooking by the project activity are then calculated as follows:

$$TR_y = (T_{b,y} - T_{p,y}) / T_{b,y}$$

Where:

TR_y	Total reduction in time spent on cooking and firewood collection activities
$T_{b,y}$	Baseline time spent on cooking and firewood collection activities per household per day
$T_{p,y}$	Project time spent on cooking and firewood collection activities per household per day

SDG 7 (Affordable and Clean Energy)

The VPA involves the distribution of ICS, a fuel-efficient technology, to replace the inefficient baseline cooking technologies.

To measure 7.1 (universal access to affordable, reliable and modern energy services), indicator 7.1.2 provides for the measurement of the 'proportion of the population with primary reliance on clean fuels and technology'. The outcome for SDG 7 is quantified as the additional number of persons having access to efficient cookstove technology in the project activity and is calculated using parameters from the project scenario.

$$ICS_{a,y} = ICS_{t,y} * U_{p,y}$$

Where:

$ICS_{a,y}$	Project ICS active in year y
$ICS_{t,y}$	Project ICS distributed in year y
$U_{p,y}$	Usage rate

SDG 13 (Climate Action)

The contribution to SDG 13 will be measured through carbon credits issued following the Gold Standard TPDDTEC v3.1 methodology.

Baseline Emissions

In line with application of the default factor in calculating baseline fuel consumption, as the per "Case of a Single Sample Test" section, page 22 of TPDDTEC v3.1. The statistics analysis is performed on a single data set obtained from performance field tests in the project scenario, and the baseline emission calculations are conducted as follows:

$$BE_{b,y} = B_{b,y} * ((f_{NRB} * EF_{b,fuel, CO2}) + EF_{b,fuel, nonCO2}) * NCV_{b, fuel} \quad (3)$$

Where:

$BE_{b,y}$	Emissions for baseline scenario b during the year y in tCO ₂ e
$B_{b,y}$	Quantity of fuel consumed in baseline scenario b during year y, in tons, as per by-default factors (cases with project performance field test only)
f_{NRB}	Fraction of biomass used during year y for the considered scenario that can be established as non-renewable biomass
$NCV_{b,fuel}$	Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.015 TJ/ton)
$EF_{b,fuel,CO2}$	CO ₂ emission factor of the fuel that is substituted or reduced. 112 tCO ₂ /TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel
$EF_{b,fuel,nonCO2}$	Non-CO ₂ emission factor of the fuel that is substituted or reduced

$$B_{b,y} = N_{p,y} * P_{b,y} \quad (4)$$

Where:

$N_{p,y}$	Project technology-days in the project database for project scenario p through year y
$P_{b,y}$	Specific fuel consumption for an individual technology in baseline scenario b during year y converted to tons/day

Project Emissions

$$PE_{p,y} = B_{p,y} * ((f_{NRB} * EF_{p,fuel, CO2}) + EF_{p,fuel, nonCO2}) * NCV_{p,fuel} \quad (5)$$

Where:

$PE_{p,y}$	Emissions for project scenario p during year y in tCO ₂ e
$B_{p,y}$	Quantity of fuel consumed in project scenario p during year y, in tons
f_{NRB}	Fraction of biomass used during year y that can be established as non-renewable biomass
$NCV_{p,fuel}$	Net calorific value of the project fuel (IPCC default for wood fuel, 0.015 TJ/ton). This is equal to the baseline fuel NCV in projects which use the same fuel.
$EF_{p,fuel,CO2}$	CO ₂ emission factor of the project fuel. This is equal to the baseline fuel ER in projects which use the same fuel, 112 tCO ₂ /TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel
$EF_{p,fuel,nonCO2}$	Non-CO ₂ emission factor of the project fuel. This is equal to the baseline fuel in projects which use the same fuel.

$$\text{Where } B_{p,y} = N_{p,y} * ((P_{p,y} * U_{p,y}) + (P_{b,y} * (1 - U_{p,y})))$$

Where

$N_{p,y}$	Project technology-days in the project database for project scenario p through year y
$P_{p,y}$	Specific fuel consumption for an individual technology in project scenario p during year y converted to tons/day
$P_{b,y}$	Specific fuel consumption for an individual technology in baseline scenario b during year y converted to tons/day
$U_{p,y}$	Cumulative usage rate for technologies in project scenario j during year y, based on cumulative installation rate and drop-off rate

Emission Reduction Calculation

$$ER_y = \sum BE_{b,y} - \sum PE_{p,y} - \sum LE_{p,y} \quad (7)$$

Where:

ER _y	Emission reduction for total project activity in year y (tCO ₂ e/yr)
BE _{p,y}	Baseline emissions for baseline scenario b in year y (tCO ₂ e/yr)
PE _{b,y}	Project emissions for project scenario p in year y (tCO ₂ e/yr)
LE _{p,y}	Leakage for project scenario p in year y (tCO ₂ e/yr)

B.6.2. Data and parameters fixed ex ante

SDG 13

Data/parameter	EF _{b,fuel,CO2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of wood-fuel in baseline scenario (includes production, transport and use)
Source of data	Calculated from IPCC defaults Volume 2: 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 2, Table 2.5
Value(s) applied	112
Choice of data or Measurement methods and procedures	Deemed valid by Methodology.
Purpose of data	Calculation of baseline emissions.
Additional comment	
	-

Data/parameter	EF _{b,non co2}
Unit	tCO ₂ e/TJ
Description	Non-CO ₂ (CH ₄ and N ₂ O) emission factor arising from use of wood fuel in baseline scenario
Source of data	IPCC Default emissions factor

Value(s) applied	8.692 (Pre-2021) 9.46 (2021 Onwards)Will be reported as weighted average for each MP
Choice of data or Measurement methods and procedures	Deemed valid by Methodology
Purpose of data	Calculation of emission reductions.
Additional comment	-

Data/parameter	EF _{p,co2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of wood fuel in project scenario
Source of data	Volume 2: 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 2, Table 2.5
Value(s) applied	112
Choice of data or Measurement methods and procedures	Deemed valid by Methodology.
Purpose of data	Calculation of emission reductions.
Additional comment	-

Data/parameter	EF _{p,non co2}
Unit	tCO _{2e} /TJ
Description	Non-CO ₂ (CH ₄ and N ₂ O) emission factor arising from use of wood fuel in project scenario
Source of data	IPCC Default emissions factor
Value(s) applied	8.692 (Pre-2021) 9.46(2021 Onwards) Will be reported as weighted average for each MP

Choice of data or Measurement methods and procedures	Deemed valid by Methodology
Purpose of data	Calculation of emission reductions.
Additional comment	-

Data/parameter	NCV _{fuel}
Unit	TJ/ton
Description	Net calorific value of the wood fuel used in the baseline
Source of data	TPPTDEC v3.1
Value(s) applied	0.015
Choice of data or Measurement methods and procedures	Deemed valid by Methodology.
Purpose of data	Calculation of emission reductions.
Additional comment	-

Data/parameter	Weighting _{3SF}
Unit	%
Description	Percentage of baseline scenario that is comprised of three stone fire (3sf) cooking technology
Source of data	Baseline Kitchen Survey
Value(s) applied	85.98%
Choice of data or Measurement methods and procedures	Measured by question in the baseline kitchen survey.
Purpose of data	Calculation of emission reductions.

Additional comment	-
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Data/parameter	Weighting _{ICS}
Unit	%
Description	Percentage of baseline scenario that is comprised of three Improved Cookstove (ICS) technology
Source of data	Baseline Kitchen Survey
Value(s) applied	14.02%
Choice of data or Measurement methods and procedures	Measured by question in the baseline kitchen survey.
Purpose of data	Calculation of emission reductions.
Additional comment	Inclusive of Improved Charcoal Stoves (see. Section B.4. for details)

Data/parameter	$P_{b,y}$
Unit	kg/household/day
Description	Quantity of fuel that is consumed in the baseline scenario b during year y (kg/household-day)
Source of data	$((\text{Default Factor} * \text{Number of People per Household}) * 1000) / 365$ $(0.5 * 4.4) * 1000 / 365$
Value(s) applied	6.03
Choice of data or Measurement methods and procedures	Deemed valid by Methodology.
Purpose of data	Calculating quantity of fuel used in the baseline scenario.
Additional comment	Methodology Default Value applied.

	The reference for number of people per household has been sourced from the 2018 Malawi Population and Housing Census Main Report. The average household size within Central and Southern region (respectively 4.4 and 4.3 people) is 4.4. Pages 31+ 32, Figure 4.1: Average household size by region, 2008 and 2018.
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SDG 5

Data/parameter	$T_{b,y}$
Unit	Hours
Description	Time spent on cooking and firewood collection activities per household per week prior to project in year y
Source of data	Baseline Kitchen Survey questions 28, 33, 35
Value(s) applied	7.76
Choice of data or Measurement methods and procedures	Measured by question in the baseline kitchen survey.
Purpose of data	Calculating time saved collecting firewood and cooking by project.
Additional comment	-

B.6.3. Ex ante estimation of SDG Impact

SDG 3: Good Health and Well Being	Description	$HAPR_y = ((P_{b,y} - P_{p,y})/P_{b,y}) * U_{p,y}$	Unit
HAPR _y	Total reduction in Household Air Pollution for project activity in year y (%)	31%	%
P _{b,y}	Quantity of fuel that is consumed in the baseline scenario b during year y (kg/household-day)	6.16	kg/hh
P _{p,y}	Quantity of fuel that is consumed in the project scenario p during year y (kg/household-day)	3.77	kg/hh
U _{p,y}	Usage rate in project scenario p during year y	80%	%

SDG 5: Gender Equality	Description	$TR_y = (T_{b,y} - T_{p,y}) / T_{b,y}$	Unit
TR _y	Total reduction time spent collecting firewood and cooking for project activity in year y	40%	%
T _{b,y}	Baseline time spent collecting firewood and cooking per household per day	7.76	hours
T _{p,y}	Project time spent collecting firewood and cooking per household per day	4.66	hours

SDG 7: Affordable and Clean Energy	Description	$N_{p,y} * U_{p,y}$	Unit
Active ICS disseminated by project	The number of active ICS disseminated by the project	24,000	Number
N _{p,y}	Number of stoves distributed	30,000	Number
U _{p,y}	Usage rate in project scenario during year y	80%	%

SDG 13: Climate Action	Description		Unit
Er _y	CO2 emission reductions per year	34,034	VER

Breakdown of SDG calculation for 1 full year crediting (see ER Calcs for reference of values);

Baseline Emissions

$$BE_{b,y} = B_{b,y} * ((f_{NRB} * EF_{b,fuel, CO2}) + EF_{b,fuel, nonCO2}) * NCV_{b, fuel} \quad (3)$$

$$113,5934 = 46,22500 * ((0.94 * 112) + 9.46) * 0.015$$

$$\text{Where } B_{b,y} = N_{p,y} * P_{b,y} \quad (4)$$

$$66,000 = 109,500,00 * 0.006027397$$

Project Emissions

$$PE_{p,y} = B_{p,y} * ((f_{NRB} * EF_{p,fuel, CO2}) + EF_{p,fuel, nonCO2}) * NCV_{p,fuel} \quad (5)$$

$$79,558 = 46,225 * ((0.94 * 112) + 9.46) * 0.015$$

$$\text{Where } B_{p,y} = N_{p,y} * ((P_{p,y} * U_{p,y}) + (P_{b,y} * (1 - U_{p,y}))) \quad (6)$$

$$46,225 = 10950000 * ((0.00377 * 0.8) + (0.006027397 * (1 - 0.8)))$$

Emission Reduction Calculation

$$ER_y = \sum BE_{b,y} - \sum PE_{p,y} - \sum LE_{p,y} \quad (7)$$

$$34,034 = 113,593 - 79,558 - 0$$

34,034

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 3

Year	Baseline estimate	Project estimate	Net benefit
2020 – 16/06/2020 – 31/12/2020	0% reduction in exposure to Household Air Pollution	31% reduction in exposure to Household Air Pollution	31% reduction in exposure to Household Air Pollution
2021	0%	31%	31%
2022	0%	31%	31%
2023	0%	31%	31%
2024	0%	31%	31%
2025 – 01/01/2025 – 15/06/2025	0%	31%	31%
Total	0%	31%	31%
Total number of crediting years	5		
Annual average over the crediting period	0% reduction in exposure to Household Air Pollution	31% reduction in exposure to Household Air Pollution	31% reduction in exposure to Household Air Pollution

SDG 5

Year	Baseline estimate	Project estimate	Net benefit
2020 – 16/06/2020 – 31/12/2020	0% reduction in time spent cooking and collecting firewood	40% reduction in time spent cooking and collecting firewood	40% reduction in time spent cooking and collecting firewood

2021	0%	40%	40%
2022	0%	40%	40%
2023	0%	40%	40%
2024	0%	40%	40%
2025 – 01/01/2025 – 15/06/2025	0%	40%	40%
Total	0%	40%	40%
Total number of crediting years	5		
Annual average over the crediting period	0% reduction in time spent cooking and collecting firewood	40% reduction in time spent cooking and collecting firewood	40% reduction in time spent cooking and collecting firewood

SDG 7

Year	Baseline estimate	Project estimate	Net benefit
2020 – 16/06/2020 – 31/12/2020	0 active ICS	24,000 active ICS	24,000 active ICS
2021	0	24,000	24,000
2022	0	24,000	24,000
2023	0	24,000	24,000
2024	0	24,000	24,000
2025 – 01/01/2025 – 15/06/2025	0	24,000	24,000
Total	0	24,000	24,000
Total number of crediting years	5		
Annual average over the crediting period	0 active ICS	24,000 active ICS	24,000 Active ICS

SDG 13

Year ¹²	Baseline estimate	Project estimate	Net benefit
2020 – 17/06/2020 – 31/12/2020	60,899 tCO ₂ e	42,652 tCO ₂ e	18,246 tCO ₂ e
2021	113,593	79,558	34,034
2022	113,593	79,558	34,034
2023	113,593	79,558	34,034
2024	113,593	79,558	34,034
2025 – 01/01/2025 – 16/06/2025	51,661	36,183	15,479
Total	566,932	397,067	169,861
Total number of crediting years	5		
Annual average over the crediting period	113,593 tCO₂e	79,558 tCO₂e	34,034 tCO₂e

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13

Data / Parameter	Np,y
Unit	Project Technologies Credited (units)
Description	Technologies in the project database for project scenario p through year y
Source of data	Cookstove Database

¹² The Crediting Period is from 01/05/2021 – 30/04/2026. Years 1 and 6 are therefore not full 365-day periods. The 2021 and 2025 vintages has respective tabs in the Ex-antes document to transparently show calculation for these.

Value(s) applied	To be monitored
Measurement methods and procedures	Sum of the total number of cookstoves distributed multiplied by the number of days crediting each technology earns in a given monitoring period
Monitoring frequency	Continuous recording
QA/QC procedures	Sales records are analysed prior to calculation of parameter
Purpose of data	Calculation of emission reductions
Additional comment	-

Data / Parameter	$U_{p,y}$
Unit	Percentage
Description	Usage rate in project scenario p through year y
Source of data	Usage Survey
Value(s) applied	Estimated at 80%. To be monitored
Measurement methods and procedures	Usage Survey
Monitoring frequency	Measured annually
QA/QC procedures	The usage survey will be carried out by trained local staff to meet the specific requirements of the methodology. All data presented in excel is subject to checking and cross referencing of a sample of the raw data by CO2balance
Purpose of data	Calculation of usage
Additional comment	-

Data / Parameter	$f_{NRB,i,y}$
Unit	Fractional non-renewability
Description	Non-renewability status of woody biomass fuel in scenario i during year y
Source of data	The NRB assessment has been carried out by an independent consultant C4 Ecosolutions in accordance with the methodology.
Value(s) applied	0.94

Measurement methods and procedures	NRB assessment by independent contractor C4 Ecosolutions
Monitoring frequency	Measured Annually
QA/QC procedures	
Purpose of data	Calculation of emission reductions
Additional comment	-

Data / Parameter	LE _{p,y}
Unit	tCO ₂ e per year
Description	Leakage in project scenario p during year y
Source of data	Monitoring Kitchen Surveys supported by desk-based research (leakage assessment)
Value(s) applied	Estimated at 0. To be monitored
Measurement methods and procedures	Leakage Assessment
Monitoring frequency	Leakage factors are re-assessed biennially.
QA/QC procedures	Research and surveys checked by senior project staff.
Purpose of data	Calculation of leakage.
Additional comment	-

Data / Parameter	P _{p,y}
Unit	kg/household/day
Description	Quantity of fuel that is consumed in the project scenario p during year y (kg/household)
Source of data	Kitchen Performance Test
Value(s) applied	To be monitored Current Value: 3.77
Measurement methods and procedures	Calculating quantity of fuel used in the project scenario using the KPT survey
Monitoring frequency	Measured biennially
QA/QC procedures	

Purpose of data	Calculation of SDG 3
Additional comment	

SDG 3

Data / Parameter	HAPR,y
Unit	percentage
Description	Total reduction in Household Air Pollution for project activity in year y (%)
Source of data	Project kitchen Survey and baseline kitchen survey
Value(s) applied	To be monitored
Measurement methods and procedures	The average difference in fuel use per technology per day in the project will be subtracted from the baseline figure. This difference will then be used to calculate the % decrease in fuel use, which is the figure used to calculate the VPAs' contribution to the SDG target
Monitoring frequency	Measured annually
QA/QC procedures	
Purpose of data	To calculate the overall reduction in household air pollution from the project activity compared to the baseline scenario
Additional comment	-

SDG 5

Data / Parameter	TP,y
Unit	Hours
Description	Time spent collecting firewood and cooking per household per week in the project in year y
Source of data	Monitoring Kitchen Survey
Value(s) applied	To be monitored
Measurement methods and procedures	Assessed every year using Project Survey
Monitoring frequency	Measured annually
QA/QC procedures	

Purpose of data	Calculation of SDG 5
Additional comment	-

Data / Parameter	TR,y
Unit	hours
Description	Total reduction in time spent collecting firewood and cooking per household in the project activity in year y
Source of data	Project Survey and baseline survey.
Value(s) applied	To be monitored
Measurement methods and procedures	Assessed every year using Project Survey. Calculate the average amount of time spent collecting firewood in the project scenario and compare to the pre-project scenario
Monitoring frequency	Measured annually
QA/QC procedures	
Purpose of data	To quantify whether the project has contributed to a reduction in the amount of time spent collecting firewood compared to the pre-project scenario towards SDG 5 target.
Additional comment	-

SDG 7

Data / Parameter	N _p ,
Unit	Number of stoves
Description	Number of stoves distributed during year y
Source of data	Sales Record
Value(s) applied	Estimated at 30,000 in Year 1, 0 thereafter. To be monitored
Measurement methods and procedures	The Sales Record of the Project will be updated at least annually.
Monitoring frequency	Annually
QA/QC procedures	CO2balance will request supporting evidence of sales record from VPA Implementer, United Purpose

Purpose of data	To determine number of Active ICS disseminated by project
Additional comment	-

B.7.2. Sampling plan

Cross sampling of devices will be applied across all homogenous VPAs in the project area. Homogenous VPAs are defined as those that are sharing a common baseline. The number of districts in the project area that will need to be sampled for a 90/30 confidence/precision will be determined. From the determined number of districts, a village from each district will be randomly selected and project households will be randomly sampled, complying with the minimum sample size for the particular survey/test.

Individual participants will be randomly selected from the cookstove electronic database. Sample sizes will be in line with the Gold Standard requirements. The random sample group is reselected for every monitoring period to ensure the selection remains random.

The surveys below will be monitored under the cross-sampling approach:

- Project Surveys- Completed annually
- Usage Surveys- Completed annually
- Kitchen Performance Test- Completed biennially

The surveys will be conducted so as to ensure that they are within the end date of the respective monitoring periods for each VPA.

B.7.3. Other elements of monitoring plan

This VPA, and any future homogenous VPAs, will be filled with improved stoves sold under the project, each with a unique identification number imprinted to the stove. Purchasers of the stove will be informed about the carbon transfer with the terms set out in the warranty receipt.

Sales Record

During the sales process, representatives of United Purpose will collect data and educate recipients on proper usage of the technology.

A comprehensive Sales Record will record the following information and will be backed up electronically. The representatives will endeavor to collect all of the following information in order to build a comprehensive Sales Record. Section 3.1.A page 29 (footnote 31) of TPDDTEC v3.1 provides that details of end users are not required in cases it is *"not feasible"* to collect them, in particular cases where *"sales of small items (such as portable cook stoves, heat retention cookers, water filters) sold in market stalls or shops where the retailer cannot reasonably be expected to collect customers names and addresses during busy times"*. Due to the nature of stove dissemination in this Project, this caveat may be relied upon where it is not feasible for the vendor to collect all information. Due to the volume of sales expected (3000 units) and the training of LMDs, the Sales Record satisfy the size requirement to *"ensure an adequate end user pool to which random sampling can be applied"*. It is in the interest of the vendor to collect the end user information, including telephone number, as each user receives follow up.

The following data will be collected:

- Date of sale
- Geographic area of sale
- Model/type of project technology sold
- Quantity of project technologies sold
- Name, address and telephone number of end user (where feasible)
- Unique stove ID

Carbon Transfer Form

The vendee will be informed about the transfer of carbon rights at the point of sale. Each vendee receives a warranty receipt which again states the terms of the transfer of carbon rights.

The details will be added to the electronic database which will provide the value for PTDs and be used to randomly select the monitoring samples.

Project Database

The Project Database is derived from the Sales Record. The Database tracks the active stoves which provides the value for PTDs and is used to randomly select the monitoring samples.

Ongoing Monitoring Studies

All monitoring studies will be recorded on a hard copy which the monitoring team will transfer into digital format for analysis documentation and reporting by United Purpose. The hard copy is scanned and then filed to enable transparent auditing.

a) Monitoring Kitchen Survey

- Completed annually.
- Begins within 1 year after project registration.
- Surveys end users currently using project technologies to explore changes in project scenario over time.
- Sample size guidance same as Baseline Survey:
 - Group Size <300: Minimum sample size 30 or population size, whichever is smaller.
 - Group Size 300 – 1000: Minimum sample size 10% of group size
 - Group Size > 1000: Minimum sample size 100

b) Usage Survey

- Completed annually
- Provides a usage rate

Usage Surveys will be carried out in line with the Good Practice Monitoring Requirements included in the Requirements and Guidelines for carrying out usage surveys for projects implementing improved cooking devices Rule Update, including:

- Kitchen observation – usage surveys will be carried out in each household in order to gather objective information.
- Interview with the primary cook – each usage survey interview will be carried out to gather information on stove use patterns
- Photos of the cooking area(s) – a photograph will be taken of the cooking area at the end of each usage survey upon consent of the interviewee
- GPS co-ordinates – GPS co-ordinates will be gathered during each usage survey
- Field team training and supervision – training will be conducted annually with the field team before the surveys are carried out. Data will be reviewed regularly as it is collected to ensure that the expected standard was maintained
- End-User Training, follow up visits and Awareness Campaign – demonstrations and training take place at the point of sale. All sales are followed up with a phone call to collect the users' feedback and advise of stove usage
- Verification checks- 5-10% of surveys will be verified by phone after the monitoring has been conducted

In line with the Good Practice Monitoring Requirements the usage rate will be capped at 90% in each monitoring period.

c) Project FT Update

- Completed biennially and in time for first verification.
- Extension of the KPT to provide fuel consumption representation of project technologies currently in use.
- Carried out by United Purpose under the supervision of CO2balance.
- Guidance on the sampling strategy will be provided by CO2balance.

d) Leakage Assessment

- Completed biennially.
- The potential sources of leakage will be investigated. If the assessment quantifies an increase in fuel consumption by the non-project households attributable to the project activity, then calculations will be adjusted to account for this.

e) Project Technology Days

- The number of project technology days is a measure of the number of technologies sold and the number of days each stove has been operating within a crediting period.
- The total number of project technology days will be calculated by using the date of each individual stove installation completion, indicated by the date of signed Carbon Transfer Form, to determine the number of days it operated within a crediting period. The profile for each individual stove within the project will then be summed to derive the total number of project technology days.

DOE Validation and Verification Site Visits

An independent Designated Operational Entity (DOE) will be appointed to review the Project and conduct a site visit as part of the Validation and Verification processes. As this VPA forms part of a group of homogenous VPAs, a DOE site visit at Validation and Verification is only required for the first VPA in the group. Subsequent homogenous VPAs that are Included or Verified under the PoA will undergo a Compliance Check to ensure that they meet the criteria provided in the PoA-DD.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of project

16/06/2020 (One year prior to first submission date 15/06/2021)

C.1.2. Expected operational lifetime of project

15 years (5 years twice renewable)

C.2. Crediting period of project

C.2.1. Start date of crediting period

17/06/2020

C.2.2. Total length of crediting period

5 years twice renewable

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 3. Community	
Health, Safety and Working Conditions	Will be measured through SDG 3.
Principle 5. Corruption	The communities will be able to communicate any cases of corruption through the continuous input mechanism. The continuous input mechanism will be monitored and any reports of corruption will be acted on.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project will reduce the amount of resources required to cook. There will be less burden on women, men and children, as less firewood for cooking needs to be collected. This will reduce the time burden on women and men in the socially isolating activity of collecting resources.
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	Equal participation of women and men in decision making is encouraged by promoting their equal membership on water point committees. These WPCs are trained to facilitate the participation of members depending on their specific circumstances. They also assist all community members to provide feedback on the project, regardless of their situation. Both women and men will benefit from the project activities, no group is excluded from participating in the project activities or use of the stoves. The project will decrease the workload of women in collecting firewood and time spent cooking, thereby allowing more time to engage in other activities. The project aims to benefit the whole community equally and women's equal participation in the LSC is encouraged. The Project will increase natural protection as no firewood will be required for cooking. No further risks or hazards for women were identified.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The Republic Of Malawi National Gender Policy, Second Edition, 2015, details Malawi's Policy on Gender and Development. These policy documents will be used as a guide to align the project with current national gender policies and best practices in Malawi. Some of the main objectives of these policies and how the project aligns with these are as follows: Objective 1: "To advocate for increased access, retention and completion to quality education for girls and boys."¹³

¹³ The Republic Of Malawi, National Gender Policy, Second Edition, 2015
https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&ved=2ahUKFwjDhr213pnxAhWN-KOKHeyPDVsQFjABegQIBRAD&url=https%3A%2F%2Fnpc.mw%2Fwp-content%2Fuploads%2F2020%2F07%2FNational-Gender-Policy-2015.pdf&usg=AOvVaw3tSvXEYI8VO_6sGy1y3fQs

The project seeks to promote gender equality in all project related activities such as fuel collection, plant management, and best WASH practices. The project aims to impact upon sustainable development within the communities it touches, including contributing to the sustainable development goals, one of which is SDG 5 gender equality. The impact of the project on SDG 5 will be measured on an annual basis during project monitoring to determine the contribution towards the SDG 5 target 5.4 - Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate.

Objective 4 "To reduce poverty among women and other vulnerable groups (Orphans, widows, PLHA, persons with disabilities, the elderly) through economic empowerment."

The Safe improved cookstoves will reduce the burden on both women and men of to collect firewood, additionally women in particular benefit from reduce time cooking and exposure to smoke. This gives them more time and opportunity to pursue development enhancing activities such as income generating activities and education, which may have once been sacrificed to provide firewood for the homestead. This acts to close the gender gap by allowing women and girls more time for self-improvement and development. The project survey will monitor the impacts of time saved from the project and what activities respondents are choosing to do with their time saved. This helps beneficiaries achieve a more sustainable livelihood over the project lifetime.

Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

An initial remote stakeholder consultation was conducted, followed by an in person event when able due to COVID restrictions, as detailed in the SCR.

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	Feedback books will be given to the Traditional Authority in each District that the project is operating in. Each book will contain the phone number for UPs Energy Projects Coordinator and contact details of CO2balance so that stakeholders can notify UP that there is written feedback needing attention.
GS Contact (mandatory)	help@goldstandard.org
Telephone access	Local stakeholders will be able to call the United Purpose Office locally on 01 822 705 or 01 823 761 and request to speak to the Energy Projects Coordinator to discuss issues with the project.
Internet/email access	Local stakeholders with access to the internet will be able to write to the UP Energy Programme

Manager and CO2balance using the following email address:
 Lloyd.archer@united-purpose.org
 Emma.donnachie@co2balance.com

Other

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	The project will adhere to all human rights requirements including respecting internationally proclaimed human rights and Universal Declaration of Human Rights and will not discriminate in any way.	N/A
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead	1. No	Both women and men will benefit from the project	N/A

to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	2. No 3. No 4. No	activities, no group is excluded from participating in the project activities or use of the stoves. The project decreases the workload of women in collecting firewood and cooking, thereby allowing more time to engage in other activities. No further risks or hazards for women and girls have been identified.	N/A N/A N/A
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	Yes	The project will reduce the community exposure to illnesses associated with household air pollution through the distribution of improved cookstoves	Will be measured through SDG 3.

Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project area does not include cultural and historic sites. The focus of the project is distributing improved cookstoves only.	N/A
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The focus of the project is distributing improved cookstoves only.	N/A
>>			
Principle 4.3 Land Tenure and Other Rights			
1. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	1. No	1. No changes to land tenure arrangements and/or rights are required.	N/A
2. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access	2. No	2. No changes to land tenure arrangements	N/A

rights, usage rights or land ownership?		and/or rights are required.	
>>			
Principle 4.4 Indigenous Peoples			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	Yes	Project will not have an impact on land usage and access rights.	N/A
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	Yes	The project shall ensure that all forms of corruption are avoided. Project beneficiaries are able to contact the project developer and implementer through the continuous grievance mechanism to report any form of corruption.	The communities will be able to communicate any cases of corruption through the continuous input mechanism. The continuous input mechanism will be monitored and any reports of corruption will be acted on.
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational	No	The project adheres to all labour laws and requirements	N/A

health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision			
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for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic	No	There are no negative economic impacts and	N/A

consequences during and after project implementation?		potential risks to local economy.	
>>			
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	GHG emissions will be reduced through replacing less efficient cooking methods with improved cook stoves.	N/A
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The project will reduce consumption of biomass.	N/A
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential,	No	There will be no significant change in the volume of water consumed by the households.	N/A

lack of aquatic connectivity or water scarcity?			
>>			
Principle 8.2 Erosion and/or Water Body Instability			
Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	The focus of the project is distributing improved cookstoves only	N/A
Is the Project's area of influence susceptible to excessive erosion and/or water body instability?	Yes, 1998 State of Environment Report states that soil erosion is the primary environmental risk in Malawi ¹⁴	Project will contribute to reducing land degradation by reducing wood fuel supply. Therefore project is expected to have a positive impact of excessive erosion and/or water body instability.	N/A
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for	No	No crops or other products are produced in the project.	N/A

¹⁴ http://www.sdn.org.mw/enviro/soe_report/chapter_3.html

production of crops or other products?			
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	There will be no impact by the project to natural disasters.	N/A
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	No GMOs will be used in the project.	N/A
>>			
Principle 9.4 Release of pollutants			

Could the Project potentially result in the release of pollutants to the environment?	No	The Project reduces the amount of pollutants released into the air.	
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The Project does not deal with hazardous or non-hazardous chemicals and/or materials.	N/A
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	No pesticides and/or fertilisers are to be used in the project.	N/A
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The project reduces the consumption of firewood, therefore having a positive impact on forest conservation.	N/A
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality	No		N/A

of food available such as through crop regime alteration or export or economic incentives?		The project has no impact on modifying foods or altering crop regimes.	
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The project does not involve animal husbandry.	N/A
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project decreases the consumption of firewood having a positive impact on conserving forest ecosystems.	N/A
>>			
Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)?	No	There are several endangered species in Malawi. The project is not envisaged to have any impact on their habitat as it only affects firewood use. The project does not impact other	N/A

AND/OR Does the Project potentially impact other areas where endangered species may be present through transboundary affects?		areas where endangered species are present.	
>>			

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	CO2balance UK Ltd
Registration number with relevant authority	04903159
Street/P.O. Box	Cook Way
Building	1 Discovery House
City	Taunton
State/Region	Somerset
Postcode	TA2 6BJ
Country	United Kingdom
Telephone	+44 (0) 1823 332233
E-mail	oscar.lozada@co2balance.com
Website	www.co2balance.com
Contact person	Oscar Lozada
Title	Senior Project Manager
Salutation	Mr
Last name	Lozada
Middle name	
First name	Oscar
Department	Projects
Mobile	mailto:
Direct tel.	+44 (0) 1823 332233
Personal e-mail	oscar.lozada@co2balance.com

APPENDIX 3-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Annex A of [Principles and Requirements](#) for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption